FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

FIITJEE

JEE(Main)-2016

ALL INDIA INTEGRATED TEST SERIES

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST–V
(MAIN)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	A	A
2.	C	A	C
3.	A	A	B
4.	D	B	B
5.	A	B	A
6.	D	A	D
7.	D	C	D
8.	B	A	D
9.	B	C	B
10.	A	C	B
11.	C	A	B
12.	B	C	B
13.	C	B	C
14.	A	A	D
15.	B	C	C
16.	C	A	B
17.	B	C	C
18.	B	B	C
19.	C	B	C
20.	B	A	D
21.	C	B	A
22.	A	D	C
23.	B	A	D
24.	C	A	A
25.	C	D	B
26.	A	C	C
27.	A	B	A
28.	D	C	A
29.	D	A	D
30.	D	D	D

Physics

PART – I

SECTION – A

1. For isochoric process $P \propto T$

$\therefore \quad \frac{\Delta P}{P} = \frac{\Delta T}{T}$

$\therefore \quad T = \frac{\Delta TP}{\Delta P} = \frac{1\times 100}{0.4} = 250$ K.

2. $P = \frac{1}{2}\mu\omega^2 a^2 v$

$\omega = \frac{2\pi v}{\lambda}$ and $v = \sqrt{\frac{T}{\mu}}$

3. $P_0(\pi r^2)$ + weight of liquid above sphere.

4. $\omega' = \sqrt{\frac{K}{m} - \frac{b^2}{4m^2}}$

5. $I_{shell} + I_{element} = 0$

6. $\frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{3}\left(\frac{1}{2}m\omega^2 x^2\right)$

$\Rightarrow \quad x = \frac{\sqrt{3}}{2}A = A\sin\omega t$

$\Rightarrow t = 2/3$ sec

7. $v_{CM} = \frac{20-6}{7} = 2$ m/s …(i)

$\frac{1}{2}5(4)^2 + \frac{1}{2}(2)(3)^2 = \frac{1}{2}5v_1^2 + \frac{1}{2}2v_2^2$ …(ii)

$v_{CM} = 2 = \frac{5v_1 + 2v_2}{7}$ …(iii)

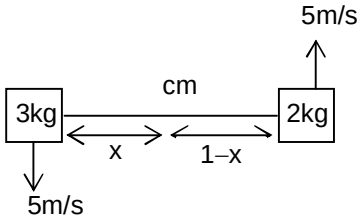
Solving (ii) and (iii), we lead to desired result.

8. From centre of mass frame

$T = \frac{mv^2}{r}$

$T = \frac{3(4)^2}{(2/5)} = 120$N , where x = 2/5

where, $v_{CM} = \frac{10-15}{5} = -1$ m/s



9. $v^2 - u^2 = kt^2/2m$

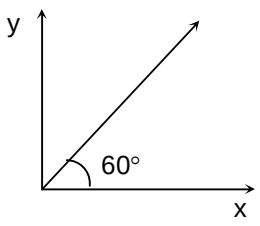
10. Momentum conservation

$12\hat{i} + 6\hat{j} = 5\vec{v}$

$\vec{v} = \frac{12\hat{i} + 6\hat{j}}{5}$ m/s

So, $\tan\theta = 6/12$

$\Rightarrow \theta = \tan^{-1}(1/2)$



$$= \left\{\frac{1}{2}\times\frac{5(144+36)}{25}\right\} - \left\{\frac{1}{2}\times 3\times 4^2 + \frac{1}{2}\times 2\times 3^2\right\} = \left(\frac{1}{10}\times 180\right) - (33) = -15\text{J}$$

11. $I = \frac{4m\ell^2}{3} + 2m\ell^2$ when $(m = \lambda\ell)$

12. $2F\cos 60° = 3g + 2g$

$2F_S\frac{1}{2} = 3g + 2g$

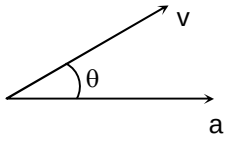
$\Rightarrow F_s = 50$

So, $a = 50/3\ m/s^2$

13. $\vec{L} = \vec{r}\times\vec{p}$

14. $a\cos\theta = \frac{|\vec{a}\cdot\vec{v}|}{v}$

$a\sin\theta = \frac{|\vec{a}\times\vec{v}|}{v}$



29 $\sqrt{\frac{\gamma_1 RT_1}{M_1}} = \sqrt{\frac{\gamma_2 RT_2}{M_2}}$

30 $P(x,t) = -F\frac{\partial y}{\partial x}\cdot\frac{\partial y}{\partial t}$

Chemistry

PART – II

SECTION – A

1. Velocity of electron in nth orbit of H – like atom is given by:

$= 2.18 \times 10^6 \times \frac{2}{n} m/s$

For second orbit of He^{2+} ion:

Z = 2, n = 2

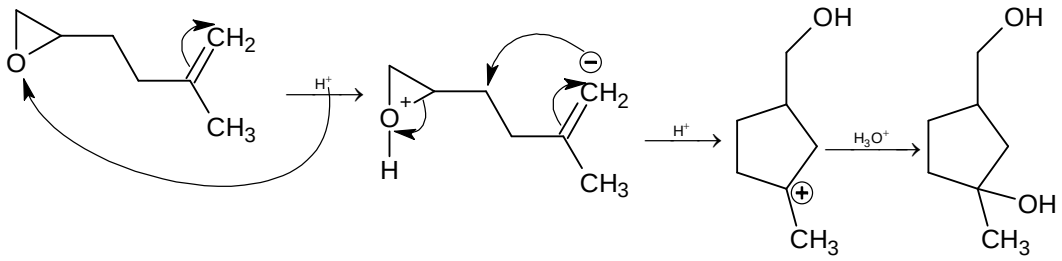
$\therefore$ Velocity of electron = 2.18×10^6 m/s

2. Cross Cannizzaro reaction will occur.

3. Percentage of $SN_2 = \frac{\text{Rate}\, SN_2}{\text{rate } SN_2 + \text{rate } SN_1} \times 100$

$= \frac{4.74 \times 10^{-5}[R-X][OH^-]}{4.74 \times 10^{-5}[R-X][OH^-] + 0.24 \times 10^{-5}[R-X]} \times 100$

= 1.93

4. 

5. $r_n = \frac{n^2 a_0}{Z}$

$\therefore r_0 \propto n^2$

$\frac{r_3}{r_1} = \frac{9}{1}$

$r_3 = 9r_1 = 9x$

$mvr_n = \frac{nh}{2\pi}$

$mv = \frac{nh}{2\pi \times r_n}$

$= \frac{3 \times h}{2\pi \times 9x} = \frac{h}{6\pi x}$

$\lambda = \frac{h}{mv} = 6\pi x$

6. Because methanoic acid contain both aldehydic as well as carboxylic group.

7. Si-F bond is stronger than C-F bond due to the donation of electron from F to Si gives rise to $p\pi$-$d\pi$ bonding.

8. N_2:

Bond order = $\frac{B.E - A.B.E}{2}$

$= \frac{10-4}{2} = 3$

CN^- is isoelectronic with N_2 & hence same B.O.

9. Bond length $= r_{Na^+} + r_{Cl^-}$

$= \frac{x}{2} + \frac{y}{2}$

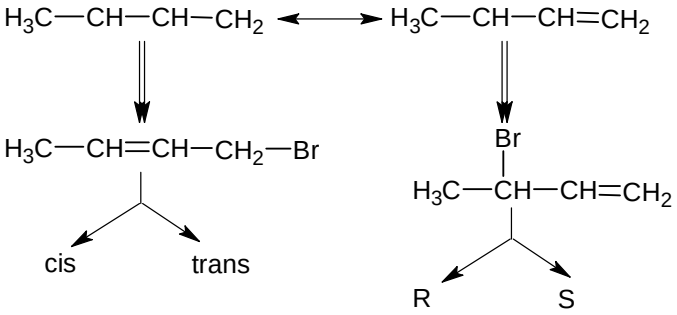
10. The distance of closest approach between Cs^+ & Cl^-.
i.e. the half of the body digonal.

$= \frac{a\sqrt{3}}{2}$

$\frac{a\sqrt{3}}{2} = 1.69 + 1.81$

$a = \frac{3.50 \times 2}{\sqrt{3}} = \frac{7}{\sqrt{3}}$

11. $CaCO_3 + SiO_2 \longrightarrow CaSiO_3 + CO_2$
flux Impurity slag

13. 

14. Except $TiCl_4$ all other metal ion Fe^{3+}, Cu^{2+} & Mn^{2+} contains unpaired electron & are coloured.

15. CdS is yellow solid.

16. $Q = i \times t$
$= 2 \times 1.93 \times 10^4$

Number of Faraday $= \frac{2 \times 1.93 \times 10^4}{96500} = 0.4F$

Number of moles of till will deposite = $\frac{23.8}{119} = 0.2$ mole

Oxidation state $Sn = \frac{0.4}{0.2} = 2$

17. In basic medium:

$K\overset{+7}{M}nO_4 \longrightarrow K_2\overset{+6}{M}nO_4$; $n - factor = 1$

Let volume strength of H_2O_2 = x ml
Now meq of $KMnO_4$ = meq of H_2O_2

$$\frac{7.9}{158/1}\times 1000 = 200\times\frac{x}{5.6}$$

$$\frac{7.9\times 1000}{158} = \frac{200x}{5.6}$$

$$x = \frac{5.6}{4} = 1.4$$

18. $N \times V_{base} = N \times V_{acid}$

$$\frac{40\times 95}{40} = \frac{39\times 100}{82/n}$$

$$n = \frac{95\times 82}{39\times 100} = 2$$

19. α is inversely porportional to concentration.

20. 1 moles of C_2H_5OH = 1 mole of HCO_2^-
= 1 × 4 eq of HCO_2^-
$\therefore$ n factor for C_2H_5OH = 4

$$\therefore E_{C_2H_5OH} = \frac{M}{4}$$

Also,
4 moles of I_2 = 1 mole of CHI_3

1 mole of $I_2 = \frac{1}{4}$ mole of CHI_3

$$= \frac{1}{4}\times 8 \text{ eq } CHI_3 = 2 \text{ eq } I_2$$

21. By using:

$$\alpha = \sqrt{\frac{k_b}{c}} = \sqrt{\left(\frac{1.85\times 10^{-5}}{0.001}\right)} = 0.136$$

Hence,

$$\left[OH^-\right] = C\alpha = 10^{-3}\times 0.136 = 1.36\times 10^{-4} M$$

$$pOH = -\log\left[OH^-\right] = -\log\left(1.36\times 10^{-4}\right)$$

= –log 1.36 + 4 = 3.86
pH + pOH = 14
pH = 14 – pOH = 14 – 3.86 = 10.14

22. Ionic compound are generally combination of metals & non metals example $CaCl_2$, while molecule compound are generally composed of non metals only as in N_2O & PCl_5.

23. $\left[Cr(H_2O)_6\right]^{3+}$ contains three unpaired electron when as $\left[Fe(CN)_6\right]^{4-}$ contains zero unpaired electron. All other complexes have one unpaired electron each.

24. The percentage of $^{10}_{5}B$ = x
Then percentage of $^{11}_{5}B$ = 100 – x

$$= \frac{10x + (100 - x)11}{100} = 10.81$$

x = 19

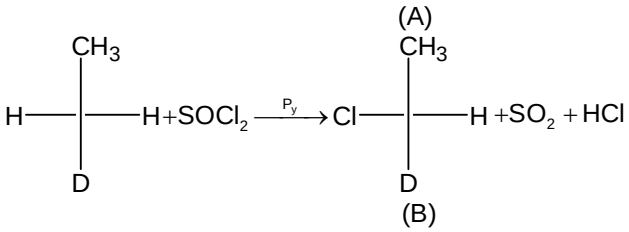
The ratio of $\frac{^{10}_{5}B}{^{11}_{5}B} = \frac{19}{81}$

25. The fact that they are non stoichiometric is related to their structure, which are similar. The CCP structure differ only in the arrangement of Fe^{2+} and Fe^{3+} in the holes in the octahedral & tetrahedral crystal.

26. Percentage of gold $\frac{22}{24} \times 100$

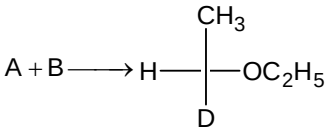
27.

$$CH_3 - CH_2 - OH + Na \longrightarrow CH_3 - CH_2 - ONa + \frac{1}{2}H_2$$



(A)

(B)



28. $K = \frac{0.693}{1.27 \times 10^{9}} = 5.46 \times 10^{-10}\ \text{year}^{-1}$

If the mass of ^{40}Ar is 3.6 times that of ^{40}K, then the original mass of ^{40}K must have been 3.6 + 1 = 4.6 times that now present.

$$t = \frac{2.303}{5.46 \times 10^{-10}\ \text{year}^{-1}} \times \log\frac{4.6}{1} = 2.8 \times 10^{9}\ \text{year}$$

30. n-factor of $Cr_2O_7^{2-}$ = 6

Equivalent of $Cr_2O_7^{2-}$ needed = equivalent of $N_2H_5^{+}$ = 0.136

$\therefore$ moles of $Cr_2O_7^{2-}$ = $\frac{0.136}{6} = 0.0227$

Mathematics

PART – III

SECTION – A

1. $\left(7-4\sqrt{3}\right)^n \to B'$ positive proper fraction

$B + B' = 1$

$\because$ $1 - B = B'$

$B'(P + B)\ \left(7+4\sqrt{3}\right)^n\left(7-4\sqrt{3}\right)^n$

$= (49 - 48) = 1.$

2. $\dfrac{3^{2n}}{8} = \dfrac{9^n}{8} = \dfrac{(1+8)^n}{8}$

$= \dfrac{{}^nC_0 + {}^nC_1 8 + {}^nC_2 8^2 \ldots {}^nC_n 8^n}{8} = \dfrac{1}{8} + \text{integer}$

$\because$ $\left\{\dfrac{3^{2n}}{8}\right\} = \dfrac{1}{8}$

3. $(1+2x)^{-1/2} = 1 - x + \dfrac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)(2x)^2}{2!} + \dfrac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)(2x)^3}{3!} \ldots$

$T_{r+1} = \dfrac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\ldots\left(-\frac{1}{2}-(r-1)\right)(2x)^r}{r!}$

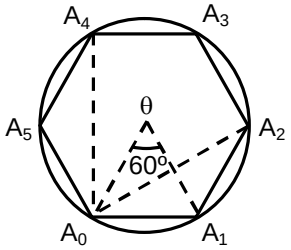
$= \dfrac{(-1)^r\left(\frac{1}{2}\right)\left(\frac{1}{2}+1\right)\left(\frac{1}{2}+2\right)\ldots\left(\frac{1}{2}+r-1\right)2^r x^r}{r!}$

$= (-1)^r \dfrac{(1.3.5\ldots(2r-1))2^r \cdot x^r}{2^r \cdot r!}$

$= (-1)^r \dfrac{(1.3.5\ldots(2r-1))x^r}{r!}$

$T_8 = (-1)^7 \dfrac{(1.3.5\ldots13)x^7}{7!}$

4. We know, in hexagon centres angle is $\dfrac{360}{6} = 60^o$ and each angle.



$= \dfrac{(2n-4)\pi}{2n} = \dfrac{(6-2)\times180^o}{6} = 120^o$

radius $r = A_0A_1$

$\therefore$ In $\Delta A_0A_1A_2$

$\cos120^o = \dfrac{A_0A_1^2 + A_1A_2^2 - A_0A_2^2}{2A_0A_1 . A_1A_2} = \dfrac{1+1-A_0A_2^2}{2.1.1}$

$A_0A_2 = \sqrt{3}$ … (i)

similar in $\Delta A_0A_5A_4$ are have $A_0A_4 = \sqrt{3}$

$(A_0A_1) \cdot (A_0A_2) \cdot (A_0A_0)$

$= \sqrt{3}\sqrt{3} = 3$ unit.

5. We have

$$\Rightarrow \quad [x] = \frac{2x\{x\}}{x+\{x\}} \Rightarrow [x] = \frac{2([x]+\{x\})\{x\}}{[x]+\{x\}+\{x\}}$$

$$\Rightarrow \quad [x] = \frac{2[x]\{x\}+2(\{x\})^2}{[x]+2\{x\}}$$

$$\Rightarrow \quad ([x])^2 + 2[x]\cdot\{x\} = 2[x]\{x\}+2(\{x\})^2$$

$$\Rightarrow \quad [x] = \sqrt{2}\{x\}$$

$$\frac{[x]}{\sqrt{2}} = \{x\}.$$

$$\Rightarrow \quad 0 \le \{x\} < 1$$

$$\Rightarrow \quad 0 \le \frac{[x]}{\sqrt{2}} < 1 \Rightarrow 0 \le [x] < \sqrt{2}$$

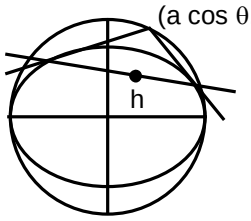
x = 0 (not possible) [x] = 0

[x] = 1, {x} = x – [x]

$$\{x\} = \frac{1}{\sqrt{2}}$$

$$x = 1+\frac{1}{\sqrt{2}} = \frac{\sqrt{2}+1}{\sqrt{2}}.$$

6. Equation of C.O.C from $(2\cos\theta, 2\sin\theta)$



C.O.C = T

$$\Rightarrow \quad \frac{2x\cos\theta}{4}+\frac{2\sin\theta y}{3} = 1 \qquad \ldots \text{(i)}$$

Let mid point $(h_1. k_1)$

equation of chord $T = s_1$

$$\frac{hx}{4}+\frac{ky}{3}-1 = \frac{h^2}{4}+\frac{k^2}{3}-1$$

$$\frac{hx}{4}+\frac{ky}{3} = \frac{h^2}{4}+\frac{k^2}{3} \qquad \ldots \text{(ii)}$$

compare both (i) & (ii)

$$\frac{2\frac{\cos\theta}{4}}{\frac{h}{4}} = \frac{\frac{2\sin\theta}{3}}{\frac{k}{3}} = \frac{1}{\frac{h^2}{4}+\frac{k^2}{3}}$$

$$\cos\theta = \frac{h\times 12}{2\left(3h^2+4k^2\right)} = \frac{6h}{\left(3h^2+4k^2\right)}$$

$$\sin\theta = \frac{12k}{\left(3h^2+4k^2\right)} = \frac{6k}{3h^2+4k^2}$$

now for locus

$36(h^2 + k^2) = (3h^2 + 4k^2)^2$

$36(x^2 + y^2) = (3x^2 + 4y^2)^2$

7. G.P. $\left\langle \begin{matrix} 1^{st}\text{ term} = a \\ \text{common ratio} = r \end{matrix} \right.$

$\sum_{i=1}^{201} ai = a + ar + ar^2 ... ar^{200}$

$625 = \frac{a\left(1-r^{201}\right)}{1-r}$ … (i)

Now,

$\sum_{i=1}^{201} \frac{1}{ai} = \left(\frac{1}{a_1} + \frac{1}{a_2} + ... \frac{1}{a_{200}} + \frac{1}{a_{201}}\right) = \left(\frac{1}{a} + \frac{1}{ar} ... \frac{1}{ar^{200}}\right)$

$= \frac{\frac{1}{a}\left(\left(\frac{1}{r}\right)^{201} - 1\right)}{\frac{1}{r} - 1} = \frac{1}{a}\left(\frac{1-r^{201}}{1-r}\right)\frac{1}{r^{200}} = \frac{1}{a} \times \frac{625}{a \cdot r^{200}}$

$= \frac{625}{\left(a_{101}\right)^2} = \frac{625}{625} = 1$

8. $x = \sec\theta - \cos\theta$

$\frac{dx}{d\theta} = \sec\theta\tan\theta + \sin\theta = \tan\theta\left(\sec\theta + \cos\theta\right)$

also $y = \sec^3\theta - \cos^3\theta$

$\frac{dy}{d\theta} = 3\sec^2\theta\sec\theta \cdot \tan\theta + 3\cos^2\theta \cdot \sin\theta$

$= 3\tan\theta\,(\sec^3\theta + \cos^3\theta)$

$\left(\frac{dy}{dx}\right)^2 = \left(\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}\right)^2 = \frac{9\tan^2\theta\left(\sec^3\theta + \cos^3\theta\right)^2}{\tan^2\theta\left(\sec\theta + \cos\theta\right)^2}$

$\left(\frac{dy}{dx}\right)^2_{x=0} \Rightarrow \left(\frac{dy}{dx}\right)^2_{\text{at } \theta = n\pi (n \in z)} = 9$

9. $bx + ay = ab$ … (i)

equation of line $\perp$ r to (i) passing through origin is

$ax - by = 0$ … (ii)

from (i) &(ii)

$a\frac{x^2}{y} + ay = ab$

$b = \frac{x^2 + y^2}{y}$ $\quad \therefore \quad a = \frac{x^2 + y^2}{x}$

Given $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$

$\frac{x^2}{\left(x^2 + y^2\right)^2} + \frac{y^2}{\left(x^2 + y^2\right)^2} = \frac{1}{c^2}$

$x^2 + y^2 = c^2$.

10. $T = s_1$ (mid point $h_1\ k_1$)

$ky - 2a(x + h) = k^2 - 4ah$

$ky - 2ax = k^2 - 2ah$

$$\frac{ky - 2ax}{k^2 - 2ah} = 1$$

parabola (homogenisation)

$y^2 = 4ax$

$$y^2 = 4ax(1) = 4a\left(\frac{ky - 2ax}{k^2 - 2ah}\right)x$$

$$y^2 + \frac{8a^2x^2}{k^2 - 2ah} - \frac{4ak}{k^2 - 2ah}xy = 0$$

angle b/w two pair of straight line

$$\tan\alpha = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\tan^2\alpha = \frac{4\left(h^2 - ab\right)}{\left(a + b\right)^2}$$

$$\tan^2\alpha = \frac{4\left[\left(\frac{2ak}{k^2 - 2ah}\right)^2 - (1)\left(\frac{8a^2}{k^2 - 2ah}\right)\right]}{\left(1 + \frac{8a^2}{k^2 - 2ah}\right)^2}$$

$(8a^2 + y^2 - 2x)^2 \tan^2\alpha = 16a^2 (4ax - y^2)$

compare with

$(32 + y^2 - 4x)^2 \tan^2\alpha = 64(8x - y)^2$

$a = 2.$

11. $(x^3 - 1) - k(x - 1) = 0$

$(x - 1)[x^2 + x + 1 - k] = 0$

$x = 1$ or $x^2 + x + (1 - k) = 0$

for exactly two distinct real solution

either $D = 0 \Rightarrow 1 - 4(1 - k) = 0 \Rightarrow 1 - 4 + 4k = 0$

$$k = \frac{3}{4}$$

or $$x^2 + x + (1 - k) = \begin{cases} 1\,(\text{one root}) \\ -2\,(\text{other root}) \end{cases}$$

$\Rightarrow 3 - k = 0 \Rightarrow k = 3.$

12. $\Delta = (a + b - c)(a + c - b)$

$\Delta = 2(s - c)2(s - b)$

$$\tan\frac{A}{2} = \sqrt{\frac{(s - b)(s - c)}{s(s - a)}} = \frac{(s - b)(s - c)}{\Delta} = \frac{1}{4}$$

$$\tan A = \frac{2\tan\frac{A}{2}}{1 - \tan^2\frac{A}{2}}$$

$$= \frac{2\times\frac{1}{4}}{1-\left(\frac{1}{4}\right)^2} = \frac{8}{15}.$$

13. $S = \frac{\tan\frac{\pi}{4}}{4} + \frac{\tan\left(\frac{3\pi}{4}\right)}{8} + \frac{\tan\left(\frac{5\pi}{4}\right)}{16} + \frac{\tan\left(\frac{7\pi}{4}\right)}{32} + \ldots$

$= \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \ldots$

$\frac{\frac{1}{4}}{1+\frac{1}{2}} = \frac{1}{4}\times\frac{2}{3} = \frac{1}{6}.$

14. Minimum values of f(x) would occur at $x = \frac{-2b}{2\left(1+b^2\right)} = \frac{-b}{1+b^2}$

$m(b) = \left(1+b^2\right)\frac{b^2}{\left(1+b^2\right)^2} - \frac{2b^2}{1+b^2} + 1$

$m(b) = \frac{-b^2}{1+b^2} + 1 = \frac{1}{1+b^2}$ varies from (0, 1] as b varies from ($-\infty$, ∞).

15. $y = x + 2$

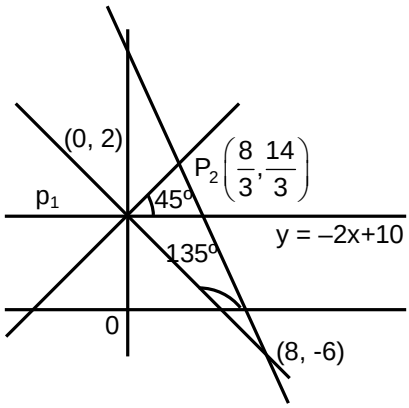
Area of triangle

$$= \frac{1}{2}\begin{vmatrix} 0 & 2 & 1 \\ \frac{8}{3} & \frac{14}{3} & 1 \\ 8 & -6 & 1 \end{vmatrix}$$

$= \frac{2}{2}\left|\frac{69}{3}\right| = \frac{64}{3}$

$\Delta = \frac{64}{3}$

$\left[\frac{64}{3}\right] = 21.$



16. $\sqrt{\log_3 x - 1} + \frac{1}{2}\frac{3\log_3 x}{-\log_3^3} + 2 > 0$

$\sqrt{\log_3 x - 1} - \frac{3}{2}\log_3 x + 2 > 0$

put $\log_3 x - 1 = t^2$ $\log_3 x = t^2 + 1$

$2|t| - 3(1 + t^2) + 4 > 0$

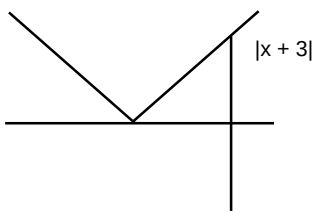
$2|t| - 3 - 3t^2 + 4 > 0$

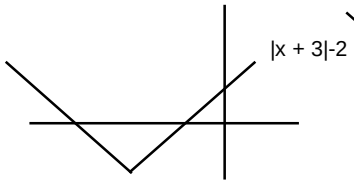
$3t^2 - 3|t| + |t| - 1 < 0$

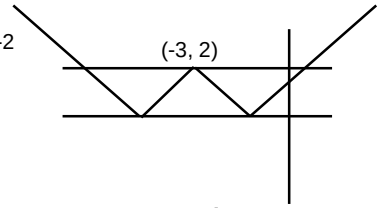
$|t| \in \left(\frac{-1}{3}, 1\right) \Rightarrow t^2 \in [0, 1)$

$x \in [3, 9) \Rightarrow$ 6 integers i.e. 3, 4, 5, 6, 7, 8.

17.







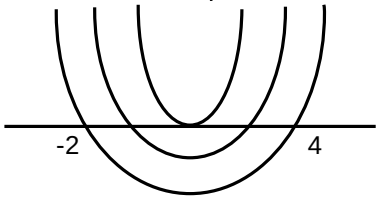
18. $A \cup B = A$

$f(x) = x^2 - ax + 4$

$D \geq 0 \Rightarrow a^2 - 16 \geq 0 \Rightarrow a \leq -4$ or $a \geq 4$ … (i)

$f(-2) \geq 0 \Rightarrow 4 + 2a + 4 \geq 0 \Rightarrow a \geq -4$ … (ii)

$f(4) \geq 0 \Rightarrow 16 - 4a + 4 \geq 0 \Rightarrow a \leq 5$ … (iii)



and $-2 \leq \frac{a}{2} \leq 4 ; -4 \leq a \leq 8$

Also, if $B = 0$ then also $A \cup B = A$

$a^2 - 16 < 0, \ a \in (-4, 4)$.

19. $\frac{\cos\theta_1}{1} = \frac{2\cos\theta_2 - 1}{2 - \cos\theta_2}$

apply C & D

$\frac{1-\cos\theta_1}{1+\cos\theta_1} = \frac{(2-\cos\theta_2)-(2\cos\theta_2-1)}{(2-\cos\theta_2)+(2\cos\theta_2-1)}$

$\tan^2\frac{\theta_1}{2} = \frac{3(1-\cos\theta_2)}{1+\cos\theta_2} = 3\tan^2\frac{\theta_2}{2}$

$\frac{\tan^2\frac{\theta_1}{2}}{\tan^2\frac{\theta_2}{2}} = 3$.

20. $x = \sqrt{y+2}$

$\left(\sqrt{y+2}\right)^5 + \left(\sqrt{y+2}\right)^2 + 1 = 0$

$y^5 + 20y^4 + 40y^3 + 79y^2 + 74y + 23 = 0$

$\therefore \quad g(x_1) \cdot g(x_2) \cdot g(x_3) \cdot g(x_4) \cdot g(x_5) =$ product of roots $= -23$

$g(x_1) \cdot g(x_2) \cdot g(x_3) \cdot g(x_4) \cdot g(x_5) - 30g(x_1x_2x_3x_4x_5)$

$= -23 + 30 = 7.$

21. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$ab + bc + ca = -\frac{1}{2}$

Now, $(ab + bc + ca)^2 = a^2b^2 + b^2c^2 + c^2a^2 + 2abc\,(a + b + c)$

$\Rightarrow \quad a^2b^2 + b^2c^2 + c^2a^2 = \frac{1}{4}$

Hence $\Rightarrow a^4 + b^4 + c^4 = (a^2 + b^2 + c^2)^2 - 2(a^2b^2 + b^2c^2 + c^2a^2)$

$= \frac{1}{2}$.

22. $\sin\left(\frac{\pi}{3}+x\right)+\sin(\alpha+x)\geq 0$

$2\sin\left(\frac{\pi}{6}+\frac{\alpha}{2}+x\right)\cdot\cos\left(\frac{\pi}{6}-\frac{\alpha}{2}\right)\geq 0$ … (i)

Now $\sin\left(\frac{\pi}{6}+\frac{\alpha}{2}+x\right)\in[-1,1]$ for $x\in R$ hence (i) will be true.

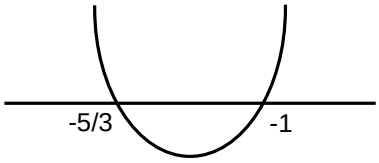
$\forall\ x\in R$ if $\frac{\pi}{6}-\frac{\alpha}{6}=(2k+1)\frac{\pi}{2}$

possible k = – 1

$\therefore\quad \frac{\alpha}{2}=\frac{\pi}{6}+\frac{\pi}{2}=\frac{2\pi}{3}$

$\alpha=\frac{4\pi}{3}=\frac{p\pi}{q}\Rightarrow p+q=7.$

23. $f'(x)=3x^2+2px+q<0\ \ x\in\left(\frac{-5}{3},-1\right)$



$3x^2+2px+q=(3x+5)(x+1)$

$2p=8;\ p=4\ \&\ q=5\Rightarrow p+q=9.$

24. $\tan\left(\frac{A-B}{2}\right)=\frac{1}{3}\tan\left(\frac{A+B}{2}\right)$ … (i)

using Napier's analogy

$\tan\left(\frac{A-B}{2}\right)=\frac{a-b}{a+b}\cot\frac{c}{2}$

$\frac{1}{3}\tan\left(\frac{A+B}{2}\right)=\frac{a-b}{a+b}\cot\frac{c}{2}$

$\frac{1}{3}\cot\frac{c}{2}=\frac{a-b}{a+b}\cot\frac{c}{2}$

a + b = 3a – 3b

a : b : : 2 : 1.

25. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=3\pi$

$\cos^{-1}x=\pi,\ \cos^{-1}y=\pi,\ \cos^{-1}z=\pi$

x = –1, y = –1, z = – 1.

Now $z^{2004}=1$

$x^{2003}=-1$

then

$$x^{2004}+y^{2004}+z^{2004}+\frac{6}{x^{2003}+y^{2003}+z^{2003}}$$

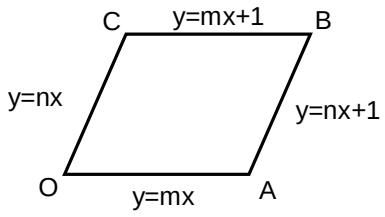
$$=1+1+1+\frac{6}{-(1+1+1)}=3-2=1.$$

26. The vertices 0(0, 0)

$A\left(\frac{1}{m-n},\frac{m}{m-n}\right)$, B(0,1)

are (parallel OABC = 2Ar(Δ OAB))

=



$$2\times\frac{1}{2}\left|0\left(\frac{m}{m-n}-1\right)+\frac{1}{m-n}(1-0)+0\left(0-\frac{m}{m-n}\right)\right|$$

$$=\frac{1}{|m-n|}.$$

27. $\angle RPS = \theta$

$\angle XPQ = 90 - \theta$

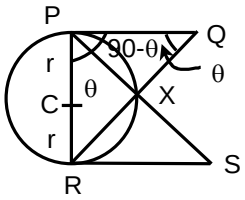
$\because$ $\angle PQX = \theta$ ($\therefore$ $\angle PXQ = 90^o$)

$\because$ ΔPRS or ΔQPR (AA similarity)

$\because$ $\frac{PR}{QP}=\frac{RS}{PR}$

$\Rightarrow$ $PR^2 = PQ \,.\, RS$

$PR=\sqrt{PQ\cdot RS} \Rightarrow 2r=\sqrt{PQ\,.\,RS}$.



28. $\therefore$ $a > 2b > 0$

Therefore, the two given circles do not intersect each other

$\therefore$ common tangents can be direct or indirect. As $y = \pm b$ are direct common tangents

$\therefore$ given line should be indirect common tangent. Clearly line $y=mx-b\sqrt{1+m^2}$ will pass from pt (a/2, 0) (mid-point of the centres of the circles)

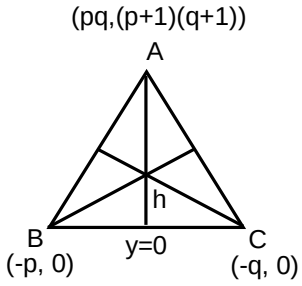
$\Rightarrow$ $m=\frac{2b}{\sqrt{a^2-4b^2}}$.

29. $AH \perp BC$ then $h = pq$

$BH \perp AC$ then $k = -pq$

$h + k = 0$

$x + y = 0$ is straight line



30. $a_1 = h_1 = 2$, $a_{10} = h_{10} = 3$

$3 = a_{10} = 2 + 9d \Rightarrow d = 1/9$

$a_4 = 2 + 3d = 7/3$

$3 = h_{10} \Rightarrow \frac{1}{3}=\frac{1}{h_{10}}=\frac{1}{2}+9D$ $\because D=-\frac{1}{54}$

$\frac{1}{h_7}=\frac{1}{2}+6D=\frac{1}{2}-\frac{1}{9}=\frac{7}{18}$

$\therefore$ $a_4\, h_7 = \frac{7}{3}\times\frac{18}{7}=6$.